higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

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AUGUST EXAMINATION

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

(8080175)

27 July 2015 (Y-Paper)
13:00–16:00

Calculators may be used.

This question paper consists of 6 pages and a formula sheet of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

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1. Answer ALL the questions
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Keep questions and subsections of questions together.
5. ALL the sketches and diagrams must be large, clear and neat.
6. Show ALL the steps and calculations.
7. Write neatly and legibly

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QUESTION 1: ALTERNATING CURRENT THEORY

1.1 Draw a neat labelled R-C integrator and indicate which type of passive filter it is. (3)

1.2 Choose the correct word(s) from those given in brackets. Write only the word(s) next to the question number (2.1.1–1.2.2) in the ANSWER BOOK.

1.2.1 In an RC-integrated circuit the mark period of the applied square waveform is (greater/smaller) than the RC-time constant.

1.2.2 (Square waveforms/Sine wave forms) indicate the presence as well as the type of error during the testing of electronic circuits.

(2 x 1) (2)

1.3 The voltage phasor diagram of a series RLC-circuit has the following values:

$$V_L = 242{,}5 \angle 38{,}32^0 \text{ V}$$

$$V_C = 164 \angle -141{,}68^0 \text{ V}$$

$$V_R = 61{,}8 \angle -51{,}68^0 \text{ V}$$

$$I_T = 5{,}15 \angle -51{,}68^0 \text{ A}$$

$$V_T = 100 \angle 0^0 \text{ V}$$

1.3.1 Draw the phasor diagram. (3)

1.3.2 Calculate:

A The reactance of the coil
B The reactance of the capacitor
C The resistance of the series resistor

(3 x 2) (6)

[14]

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QUESTION 2: POWER SUPPLIES

2.1 An RC π-filter delivers an output voltage of 12 V. If $X_{C2} = 8\ \Omega$; $R = 10\ \Omega$; $R_L = 300\ \Omega$ and $V'_{R(RMS)} = 0{,}8$ V;

Calculate:

2.1.1 V_{DC} across the first capacitor (3)

2.1.2 $V_{R(RMS)}$ across the first capacitor (3)

2.1.3 r' across the second capacitor (2)

2.2 Draw a neatly labelled circuit diagram that provides over-voltage protection in a power supply and briefly describe how the circuit operates. (6)

[14]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 The following values of a fixed biased common-emitter amplifier are known:

$+V_{cc}$	=	12 V	$I_{c(max)}$	=	20 mA	V_{ce}	= 6 V
β	=	250	$f_{(min)}$	=	100 Hz	V_{be}	= 0,6 V

Calculate:

3.1.1 R_e (3)

3.1.2 C_e (2)

3.1.3 R_c (2)

3.1.4 R_b (3)

3.2 Calculate the:

3.2.1 current gain, A_i (not necessary for the current divider rule) (4)

3.2.2 voltage gain, A_v for the circuit in Question 3.1 by means of the precision method if:

h_{ie}	=	1300 Ω	h_{re}	=	2 x 10 - 4
h_{fe}	=	250	h_{oe}	=	20 μA/V

Note: The circuit is connected to a 10 kΩ load (R_L) (3)

[17]

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QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 If the input signal of the integrator is a square wave form of ±1 V, 500 Hz and the input resistor and feedback capacitor has values of 10 MΩ and 0,01 μf respectively, calculate the output voltage and draw its waveform (capacitor is initially discharged). (9)

4.2 Draw a neatly labelled circuit diagram of a voltage follower and briefly describe TWO functions of this circuit. (4)
[13]

QUESTION 5: INTEGRATED CIRCUITS

5.1 What should be done with the unused inputs of a CMOS integrated circuit when it is connected to a circuit? (2)

5.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (5.2.1–5.2.3) in the ANSWER BOOK.

5.2.1 CMOS-integrated circuits have higher noise immunity.

5.2.2 CMOS-integrated circuits are susceptible to static charges because of their low reactive input.

5.2.3 As soon as you work on the circuit of CMOS-integrated circuits, the power supply to the circuit must be switched off.

(3 x 1) (3)
[5]

QUESTION 6: ELECTRONIC PHASE CONTROL

Draw a neatly labelled complete circuit diagram of a full-wave DC-phase control circuit for a DC load. Input and output waveforms must be shown. **[6]**

QUESTION 7: TEST EQUIPMENT

Draw a neat, labelled block diagram of a dual-slope A/D-voltmeter. **[6]**

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QUESTION 8: OSCILLATORS

8.1 Name the amount of stable and semi-stable conditions of the following circuits:

8.1.1 A-stable multivibrator

8.1.2 Mono-stable multivibrator

8.1.3 Bi-stable multivibrator

(3 x 1) (3)

8.2 Draw a neatly labelled circuit diagram of a bi-stable multivibrator, which uses two NPN transistors. (4)

8.4 Draw a neatly labelled circuit diagram of a DC-to-DC voltage converter which makes use of a 555-timer. (5)

[12]

QUESTION 9: TRANSDUCERS

9.1 Draw a neatly labelled circuit diagram of a closed-loop system which makes use of synchronous motors in order to control a radar antenna. (4)

9.2 Briefly describe what happens as soon as the transmitter rotor moves through an angle θ. (4)

9.3 Draw a neatly labelled circuit diagram of a precision switch that makes use of a light-dependent resistor and an operational amplifier. (5)

[13]

TOTAL: 100

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FORMULA SHEET: INDUSTRIAL ELECTRONICS N5

$$I = \frac{V}{R} \qquad P = IV = I^2R = \frac{V^2}{R}$$

$$V_T = V_1 + V_2 + V_3 + ... = I_1R_1 + I_2R_2 + I_3R_3 + ...$$

$$I_T = I_1 + I_2 + I_3 + ... = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + ...$$

$$T = RC \qquad T = \frac{L}{R}$$

$$V_R = RC\frac{dv}{dt} \qquad V_C = \frac{1}{RC}\int v_i dt$$

$$X_L = 2\pi fL \qquad X_C = \frac{1}{2\pi fC}$$

$$Z = R + jX_L \qquad Z = R - jX_C$$

$$Z = R + j(X_L - X_C) \qquad I_T = \frac{V_T}{Z_T}$$

$$V_R = I_T R \qquad V_L = I_T(jX_L)$$

$$V_C = I_T(-jX_C) \qquad f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{f_2 - f_1}$$

$$BW = f_2 - f_1 \qquad \frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$Z_T = \frac{Z_2Z_2}{Z_1 + Z_2} \qquad I_T = I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2}$$

$$Z_T = \frac{R(jX_L)}{R + jX_L} \qquad \frac{1}{Z_T} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L \qquad I_T = \frac{V}{R} - j\frac{V}{X_L}$$

$$Z_T = \frac{R(-jX_C)}{R - jX_C} \qquad \frac{1}{Z_T} = \frac{1}{R} + \frac{j}{X_C}$$

$$I_T = I_R + jI_C \qquad I_T = \frac{V}{R} + j\frac{V}{X_C}$$

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$$\frac{1}{Z_T} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

$$I_T = I_R - j(I_L - I_C) \qquad I_T = \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$$

$a + jb = \sqrt{a^2 + b^2} / \tan^{-1}\frac{b}{a} = r \angle \theta$

$r \angle \theta = r(\cos\theta + j\sin\theta) \qquad Q = \tan\theta$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \qquad Z_d = \frac{L}{CR_1}$$

$$V_{rms} = \frac{1}{\sqrt{2}} V_m = 0{,}707 V_m \qquad V_{dc} = \frac{2}{\pi} V_m = 0{,}637 V_m$$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P} \qquad V_{dc} = \frac{1}{\pi} V_m = 0{,}318 V_m$$

$$PIV = V_m \qquad PIV = 2V_m$$

$$R_{r(rms)} = 0{,}385 V_m \qquad V_{r(rms)} = 0{,}305 V_m$$

$$r = \frac{V_{r(rms)}}{V_{dc}} \qquad V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

$$V_{dc} = V_m - \frac{I_{dc}}{2fC} \qquad V_{dc} = V_m - \frac{I_{dc}}{4fC}$$

$$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3}fC} = \frac{V_{dc}}{2\sqrt{3}fCR_L} \qquad V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}fC} = \frac{V_{dc}}{4\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{2\sqrt{3}fCV_{dc}} = \frac{1}{2\sqrt{3}fCR_L} \qquad r = \frac{I_{dc}}{4\sqrt{3}fCV_{dc}} = \frac{1}{4\sqrt{3}fCR_L}$$

$$V'_{dc} = \frac{R_L}{R + R_L} \cdot V_{dc} \qquad V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}} \cdot V_{r(rms)}$$

$$X_C = \frac{1}{2\pi fC} \qquad X_C = \frac{1}{4\pi fC} \qquad r' = \frac{V'_{r(rms)}}{V'_{dc}}$$

$$V'_{r(rms)} = \frac{X_C}{R} \cdot V_{r(rms)} \qquad r' = rX_C\left(\frac{R + R_L}{R.R_L}\right)$$

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$$V'_{dc} = V_{dc} - I_{dc}R_1 \qquad V'_{dc} = \frac{R_L}{R_L + R_1} \cdot V_{dc}$$

$$V'_{r(rms)} = \frac{V_{r(rms)}}{(2\pi f)^2 LC} \qquad V'_{r(rms)} = \frac{V_{r(rm)}}{(4\pi f)^2 LC}$$

$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \qquad \%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$2V_m = V_{c2} = V_m + V_{c1} \qquad 3V_m = V_{c1} + V_{c3} = V_m + 2V_m$$

$$S = \frac{\Delta V_o}{\Delta V_i} \qquad V_R = V_i - V_z$$

$$R_{s(\min)} = \frac{V_{i(\max)} - V_z}{I_{z(\max)}} \qquad I_z = \frac{P_z}{V_z}$$

$$R_{L(\min)} = \frac{V_Z}{V_{i(\max)} - V_Z} \cdot R_S \qquad V_o = V_r - V_{be}$$

$$R_c = \frac{V_{cc} - V_{ce}}{I_c} \qquad R_b = \frac{V_{cc} - V_{be}}{I_b}$$

$$\beta = \frac{I_c}{I_b} \qquad C_e \geq \frac{10}{2\pi f R_e}$$

$$V_e = \frac{V_{cc}}{10} \qquad R_e = \frac{V_e}{I_e} \simeq \frac{V_e}{I_c}$$

$$R_c = \frac{V_{cc} - V_{ce} - V_e}{I_c} \qquad R_b = \frac{V_{cc} - V_{be} - V_e}{I_b}$$

$$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b} \qquad R_{b2} = \frac{1}{10}\beta R_e$$

$$V_b = V_e + V_{be}$$

$$V_{be} = h_{ie} i_b + h_{re} V_{ce} \qquad i_c = i_{fe} i_b + h_{oe} V_{ce}$$

$$A_i = \frac{h_{fe}}{1 + h_{oe} Z_L} \qquad A_i = h_{fe}$$

$$A_i = \left(\frac{h_{fe}}{1 + h_{oe} Z_L}\right)\left(\frac{R_b T}{R_{bT} + Z_1}\right)\left(\frac{R_c}{R_c + R_L}\right)$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) Z_L} \qquad A_v = \frac{-h_{fe} Z_L}{h_{ie}}$$

$$Z_1 = h_{ie} - \frac{h_{fe} h_{re} Z_L}{1 + h_{oe} Z_L} \qquad Z_1 = h_{ie}$$

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$$Z_2 = \frac{1}{h_{oe} - \frac{h_{fe}h_{re}}{h_{ie} + R_s}} \quad Z_2 = \frac{1}{h_{oe}}$$

$$A_p = \frac{A_i^2 R_L}{R_1} = -A_v A_i \qquad A_p = \frac{{h_{fe}}^2 R_L}{h_{ie}}$$

$$Z_0 = R_C // R_L // Z_2 = Z_L // Z_2$$

$$Z_0 = R_C // Z_2 = Z_L // Z_2 \qquad Z_1 = R_b // Z_1$$

$$Z_i = R_{b1} // R_{b2} // Z_1 \quad I_1 = \frac{R_{bT} I_i}{R_{bT} = Z_1}$$

$$I_0 = h_{fe} I_b = h_{fe} \left(\frac{R_{b2}(I_i)}{R_{b2} + h_{ie}} \right) \quad A_i = \frac{I_0}{I_1}$$

For common base, substitute all the 'e' subscripts with a 'b' in the h-parameters.

$$Z_L = R_c // R_L \quad I_1 = \frac{R_e I_i}{R_e + Z_1}$$

CMRR = $\frac{A_{dm}}{A_{cm}}$ *CMRR (dB) = 20log* $\frac{A_{dm}}{A_{cm}}$

$$I_e = \frac{V_e}{R_e} \qquad I_c = \frac{I_e}{2}$$

$$R_L = \frac{V_{R_L}}{I_C} \qquad g_m R_L = \frac{h_{fe}}{h_{ie}} . R_L$$

$$V_0 = -\left(\frac{R_f}{R_1} \right) . V_i \qquad V_0 = \left(\frac{R_f}{R_1} + 1 \right) . V_i$$

$$V_0 = -\left(\frac{R_f}{R_1} . V_1 + \frac{R_f}{R_2} . V_2 + \frac{R_f}{R_3} . V_3 \right) \quad V_0 = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) R_f$$

$$V_0 = -(V_1 + V_2 + V_3) \qquad V_0 = -(I_1 + I_2 + I_3)\, R_f$$

$$V_0(t) = -\frac{1}{RC} \int V_i(t) \quad V_0(t_b) = -\frac{1}{RC} \int_{t_a}^{t_b} V_i(t_b) + V_c(t_a)$$

$$t = \frac{1}{f} \quad A_v = -\frac{R_s}{R_1}$$

$$R_2 = \frac{R_1 R_s}{R_1 + R_s} \qquad f_c = \frac{1}{2\pi R_s C}$$

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$V_0(t) = -RC\frac{dV_i(t)}{dt}$ $A = \frac{R_f}{X_c}$

$A = -\frac{R_f}{R_s}$ $V_0(t) = -R_f C\frac{d}{dt}.v_i \sin\omega t$

$t = R_f C$ $V_0 = A(V_r - V_i)$

$V_0 = \frac{R_f}{R_s}(V_2 - V_1)$ $V_0 = V_2 - V_1$

$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$ $f_0 = \frac{1}{2\pi RC}$

$f_0 = \frac{1}{2\pi\sqrt{L_T C_1}}$ $L_T = L_1 + L_2 + 2M$

$f_0 = \frac{1}{2\pi\sqrt{LC_T}}$ $C_T = \frac{C_1 C_2}{C_1 + C_2}$

$f_0 = \frac{1}{2\pi\sqrt{LC_2}}$ $f = \frac{1}{2\pi RC\sqrt{6}}$

$f_0 = \frac{1{,}5}{RC}$ $f_0 = \frac{1}{t} = \frac{1}{t_1 + t_2}$

$t_1 = 0{,}7\ R_2 C_1$ $t_2 = 0{,}7\ R_1 C_2$

$f_0 = \frac{1}{1{,}4RC}$ $V_i = I_{c2}R_e + V_{be(ON)}$

$t = 1{,}1\ RC$ $f_0 = \frac{1{,}443}{(R_A + 2R_B)C}$

$t_{low} = 0{,}693(R_B)C$ $t_{high} = 0{,}693(R_A + R_B)C$

$t_T = t_{low} + t_{high}$ $K = \frac{\Delta R / R}{\Delta l / l}$

$\sigma = \Delta l / l$ $R = \rho\frac{1}{\pi d^2 / 4}$

$\sigma = \frac{S}{E}$ $\text{Resolution} = \frac{1}{amount\,of\ turns}$

$\text{Resolution} = \frac{voltage\,drop\,across\,adjacent\,turns}{total\,voltage\,drop}$

$R_t = Ae^{B/T}$ $T = 273 + °C$

$V_A = \frac{R_2}{R_1 + R_2}.V_T$ $V_B = \frac{R_t}{R_t + R_3}.V_T$

$A_v = \frac{V_0}{V_i}$

$V_{AB} = V_A - V_B$

$V_{Hall} = kIH$

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